

Datasets used in this project

All sources below are public and free — no account or API key needed for any of them. Team members can either (a) pull the small derived files already committed in `data/` after cloning this repo, or (b) hit the same links directly and regenerate everything from scratch via the scripts listed. No shared git remote is set up yet (`git remote -v` is empty) — if you want everyone on the exact same repo, someone needs to push this to a GitHub org/repo and share that URL; until then, this doc + direct links are the way to stay in sync.

1. Landslide inventory (positive samples)

GSI National Landslide Forecasting Centre (NLFC) — "Landslide Inventory (Field Validated)" - Portal:

<https://bhusanket.gsi.gov.in/> - Direct file (what we actually parsed): https://bhusanket.gsi.gov.in/pics/landslide_report.pdf -

Format: **904-page PDF, tabular** (not a CSV/shapefile — the portal's "Download Data" button just opens this PDF). We extracted it with `pdfplumber` into a real table. - Content: 36,072 records nationwide (SI.No, Slide_No, State, District, Slide_Name, NH_SH_Location, Latitude, Longitude, Material_Involved, Movement_Type, History/date) - **Already extracted for you:** [data/raw/gsi_sikkim_landslides.csv](#) — the 777-record Sikkim subset, committed in this repo. No need to re-parse the 904-page PDF yourself. - License/access: public government portal, no login, no stated usage restriction found on the site. - **Known limitation:** 96.1% of Sikkim records are within 100m of a mapped road — a field-survey artifact (see `README.md` "ML data pipeline" section). Scope your claims accordingly.

2. Elevation / terrain (DEM)

Copernicus GLO-30 Digital Elevation Model - Registry page: <https://registry.opendata.aws/copernicus-dem/> - Public

S3 bucket (no key, plain HTTPS): <https://copernicus-dem-30m.s3.amazonaws.com/> - Tile used for the Sikkim pilot:

`Copernicus_DSM_COG_10_N27_00_E088_00_DEM` — direct download:

[https://copernicus-dem-30m.s3.amazonaws.com/Copernicus_DSM_COG_10_N27_00_E088_00_DEM/Copernicus_DSM_COG_10_N27_00_E088_00_DEM/Copernicus_DSM_COG_10_N27_00_E088_00_DEM.tif](https://copernicus-dem-30m.s3.amazonaws.com/Copernicus_DSM_COG_10_N27_00_E088_00_DEM/Copernicus_DSM_COG_10_N27_00_E088_00_DEM.tif)

- Format: Cloud-Optimized GeoTIFF, ~30m resolution, native CRS EPSG:4326 (must reproject to EPSG:32645/UTM 45N before computing slope/aspect — see `scripts/ml/fetch_dem.py`) - Coverage: this single tile (27–28°N, 88–89°E) covers all 777 Sikkim inventory points — no mosaicking needed for this pilot - License: Copernicus DEM is free and open (see registry page for full terms) - **Regenerate:** `python -m scripts.ml.fetch_dem` (downloads + reprojects automatically)

(Evaluated but not used: OpenTopography's Global DEM API — requires a free account/API key, which nobody on the team has created yet. Copernicus-on-AWS needed no such step.)

3. Road network

OpenStreetMap, via the Overpass API - Query endpoint: <https://overpass-api.de/api/interpreter> - Interactive query

builder (useful for exploring by hand): <https://overpass-turbo.eu/> - Raw map browser: <https://www.openstreetmap.org/> -

Format: JSON (out `geom` Overpass query), converted to GeoJSON in our pipeline - Bounding box used: 26.9–28.2°N, 87.9–89.0°E (Sikkim + small margin) - License: **ODbL (OpenStreetMap's Open Database License)** — attribution

required if this ends up in any public-facing map/dashboard - **Regenerate:** `python -m scripts.ml.fetch_osm_roads` -

Note: the public Overpass server occasionally times out under load (hit this once this session) — just retry, or use a mirror like <https://overpass.kumi.systems/api/interpreter>

4. Rainfall

Open-Meteo — <https://open-meteo.com/> (docs: <https://open-meteo.com/en/docs>) - Forecast + recent-past endpoint

(used by the backend's live alert pipeline): <https://api.open-meteo.com/v1/forecast> - Historical archive endpoint (used for the rainfall case study, back to 2011+ confirmed, ERA5-based reanalysis, ~10km grid):

<https://archive-api.open-meteo.com/v1/archive> - No key, no login, generous free-tier rate limits - License: CC BY 4.0

(attribution required) - **Honesty note for the team/judges:** this is our fallback, not the deck's stated primary source. We

checked 6 official Government-of-Sikkim/IMD rainfall sources this session — all either unreachable (IMD Sikkim, ENVIS Sikkim — DNS failures) or had no public rainfall dataset (SSDMA, Sikkim Open Government Data portal). Don't imply IMD data is live; it isn't.

5. Rainfall intensity-duration threshold (not a dataset — a literature value)

- Harilal, G.T., Madhu, D., Ramesh, M.V. & Pullarkatt, D. (2019). "Towards establishing rainfall thresholds for a real-time landslide early warning system in Sikkim, India." *Landslides*, 16(12), 2395–2408.
- DOI: <https://doi.org/10.1007/s10346-019-01244-1>
- Configured in `.env / app/config.py` (`RainfallThresholdConfig`) — **flagged verified against primary text: false**, since the paper is paywalled and we only confirmed the coefficients via secondary summaries + Semantic Scholar metadata (<https://api.semanticscholar.org/graph/v1/paper/DOI:10.1007/s10346-019-01244-1>). If anyone on the team has journal access, verifying this directly against the primary text would be worth doing before quoting it to judges.

Checked, evaluated, and explicitly NOT used

For the record, so nobody re-spends time on these:

Source	Why not used
Itanagar Yupia Road Susceptibility Map.pdf (local file)	Arunachal Pradesh, not Sikkim; a rendered susceptibility map (raster), not extractable point/event data
Sikkim Landslide_2016.pdf (NRSC "So Bhir"/Mantam report, local file)	Real event report, but the exact same event is already in the GSI inventory (cross-validated coordinates match) — adds no new data point
IMD Sikkim (imdsikkim.gov.in)	Domain unreachable (DNS failure)
ENVIS Sikkim (sikenvvis.nic.in)	Domain unreachable (DNS failure)
Sikkim SDMA (ssdma.nic.in)	Reachable, but no public data portal — awareness/emergency-contacts site only
Sikkim Open Government Data (sikkim.data.gov.in)	Reachable, 355 catalogs, none are rainfall/meteorology
OpenTopography Global DEM API	Requires an API key/account we haven't created

Regenerating everything from scratch

```
pip install -r requirements.txt
python -m scripts.ml.fetch_dem          # DEM tile -> data/raw/dem/, data/processed/
python -m scripts.ml.fetch_osm_roads    # roads -> data/raw/sikkim_roads.geojson
python -m scripts.ml.build_negative_samples
python -m scripts.ml.build_training_dataset
python -m scripts.ml.rainfall_threshold_case_study
```

`data/raw/gsi_sikkim_landslides.csv` is already committed — there's no script to regenerate it from the 904-page PDF (that extraction takes ~5 minutes and isn't automated yet); if you need to redo it, ask and I'll add `scripts/ml/fetch_gsi_inventory.py`.